

Lesson 14 Proof by Induction

$S(n)$: statement/thm/formula about
↑
a set of integers bounded below
integer

Prop 1 For every natural $\# n$, $n \geq 1$
 $1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$

$S(1)$ is true: $n=1$

$$1 = \frac{1(1+1)}{2} \checkmark$$

$$n=2: \underbrace{1+2}_3 = \underbrace{\frac{2(2+1)}{2}}_3 \checkmark$$

Proof by induction $S(n)$, $n \geq a$

(1) Basis step (base case)
Show that $S(a)$ is true

Prop 1: $S(1)$ is true: $n=1$

$$\underline{1} = \frac{1(1+1)}{2} \checkmark$$

(2) Inductive step: Prove that for all $k \geq a$
if $S(k)$ is true, then $S(k+1)$ is true,

that is, we assume $S(k)$ is true,
and we prove that $S(k+1)$ is true.

Prop 1 We assume formula is true for
 $n=k$, (inductive hypothesis)

For $n=k+1$:

$$\begin{aligned} & 1+2+3+\dots+k+(k+1) \quad \left[\frac{(k+1)(k+2)}{2} \right] \\ &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{k+1}{2} (k+2) \end{aligned}$$

formula is true for $n=k+1$

Why induction works?

- $S(1)$ is true
- $S(k)$ is true $\rightarrow S(k+1)$ is true
 $\rightarrow S(2)$ is true
 $\rightarrow S(3)$ is true

||||| ... ||| dominos

Prop 2 $n \geq 0$

$$2^0 + 2^1 + 2^2 + \dots + 2^n = 2^{n+1} - 1$$

Basis: $n=0$

$$\underbrace{2^0}_1 \stackrel{?}{=} \underbrace{2^{0+1}}_1 - 1 \quad \checkmark$$

Inductive: Assume formula is true for $n=k$

Then for $n=k+1$,

$$\begin{aligned} & 2^0 + 2^1 + 2^2 + \dots + 2^k + 2^{k+1} \quad \underbrace{(k+1)+1}_{2^{k+2}-1} \\ = & \underbrace{2^{k+1}(-1) + 2^{k+1}}_{2^{k+1} + 2^{k+1} - 1} \\ = & 2(2^{k+1}) - 1 \\ = & \underbrace{2^{(k+1)+1}}_{2^{k+2}} - 1 \end{aligned}$$

$$a^n a^m = a^{n+m}$$

Prop 3 $n \in \mathbb{N}$

$$1 \cdot 1! + 2 \cdot 2! + \dots + n \cdot n! = \underline{(n+1)! - 1}$$

Basis: $n=1$

$$\underbrace{1 \cdot 1!}_1 \stackrel{?}{=} \underbrace{(1+1)! - 1}_{2-1} \quad \checkmark$$

Inductive: Assume formula is true for $n=k$.

Then for $n=k+1$:

$$\begin{aligned} & 1 \cdot 1! + 2 \cdot 2! + \dots + k \cdot k! + \boxed{(k+1)(k+1)!} \\ &= \underbrace{(k+1)!(-1)}_{\text{from previous step}} + \boxed{(k+1)(k+1)!} \rightarrow \\ &= (k+1)! + (k+1)(k+1)! - 1 \quad \text{with } \frac{(k+2)! - 1}{(k+1)+1} \\ &= (k+1)! (1 + (k+1)) - 1 \\ &= (k+1)! (k+2) - 1 \\ &= (k+2)! - 1 \quad \text{with } \frac{1 \cdot 2 \cdot 3 \cdot \dots \cdot (k+1)}{(k+2)} \end{aligned}$$

Prop 4 $n \in \mathbb{N}$
 $n^3 - n$ is divisible by 3

$$3 \mid (n^3 - n) \text{ or } n^3 - n = 3m, m \in \mathbb{Z}$$

Proof Basis step $n=1$

$$\underbrace{1^3 - 1}_0 \stackrel{?}{=} 3(\underbrace{0}_{\in \mathbb{Z}}) \quad \checkmark$$

Inductive : assume that

$$k^3 - k = 3m, m \in \mathbb{Z}$$

Then for $n = k+1$,

$$\begin{aligned} & (k+1)^3 - (k+1) = 3(\quad) \\ &= k^3 + 3k^2 + 3k + 1 - k - 1 \\ &= \underbrace{(k^3 - k)}_{3m} + 3k^2 + 3k \\ &= 3m + 3k^2 + 3k \\ &= 3(\underbrace{m + k^2 + k}_{\in \mathbb{Z}}) \end{aligned}$$

Statement is true for $n = k+1$

Prop 5 recursive sequence

$$a_1, a_2, a_3, \dots$$

$$\underline{a_1 = 3}, \quad a_n = 2a_{n-1} - \underline{n} + \underline{2}, \quad n \geq 2$$

$$\begin{aligned} \downarrow \\ a_2 &= 2a_1 - \cancel{2} + 2 \\ &= 2(3) = 6 \end{aligned}$$

$$a_3 = \dots$$

For all $n \in \mathbb{Z}$,

$$a_n = 2^n + n$$

Basis : $n = 1$

$$a_1 \stackrel{?}{=} \underbrace{2^1 + 1}_3 \quad \checkmark$$

Inductive: Assume that

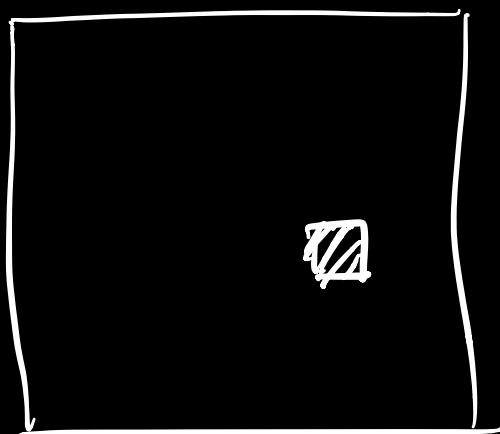
$$\boxed{a_k = 2^k + k} \quad \leftarrow$$

For $n = \underline{k+1}$:

$$a_{k+1} = 2\underline{a_k} - \underline{(k+1)} + \underline{2}$$

$$\begin{aligned}
 &= 2(2^k + k) - k - 1 + 2 \\
 &= 2(2^k) + 2k - k + 1 \\
 &= 2^{k+1} + (k+1)
 \end{aligned}$$

Prop 6 Geometric



$2^n \times 2^n$, $n \in \mathbb{Z}$
board, with
 1 square removed,
 board can be
 tiled by
 L-triominoes



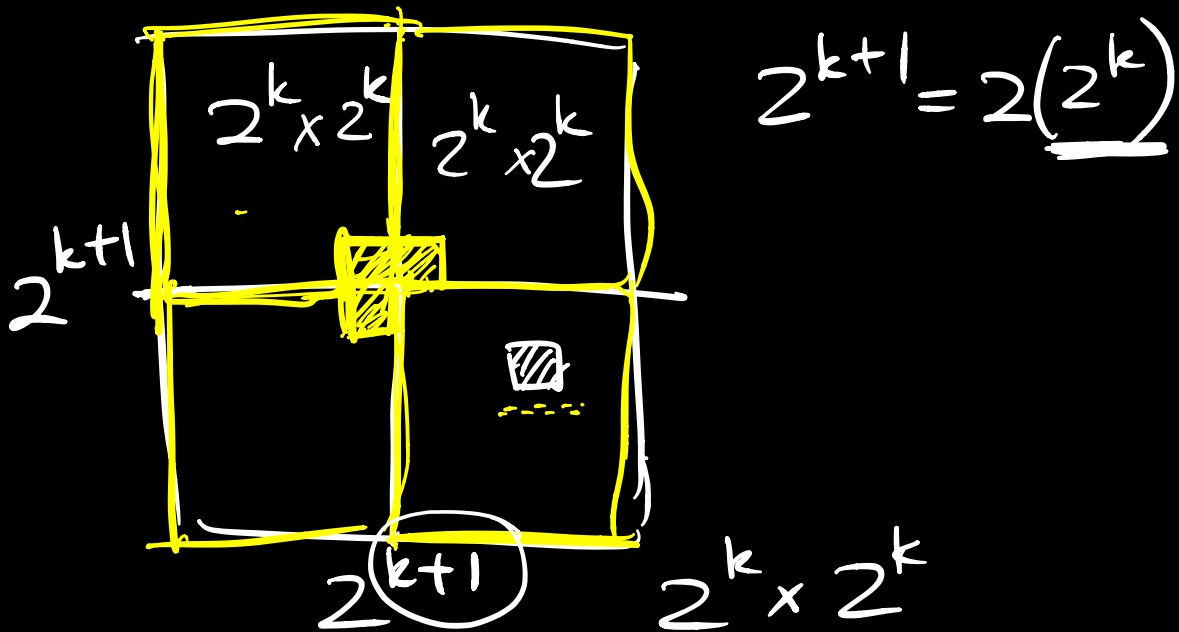
Basis: $n=1$

$$2^1 \times 2^1 = 2 \times 2$$



Inductive assume statement
is true for $2^k \times 2^k$ board

For $2^{k+1} \times 2^{k+1}$ board:



each smaller square

can be covered by L-Triominoes

So the whole board can
be tiled !